

Application Note

Design guide for Power sequence

VRender™ 3D

User Manual

MDVR3511FAN200602-V1

Ver. 1.0 / February 6, 2006

VR3511F

Application Note for Power sequence of VR3511F Processor

MAGIC EYES

This document contains information on a product under development. MagicEyes reserves the right to change or discontinue this product without notice. © MagicEyes Digital Co.ltd 2005. All rights reserved.

Important Notice

Information in this document is provided solely to enable system and software implementers to use MagicEyes products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits or integrated circuits based on the information in this document.

MagicEyes reserves the right to make changes without further notice to any products herein.

MagicEyes makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MagicEyes assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters which may be provided in MagicEyes data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. MagicEyes does not convey any license under its patent rights nor the rights of others. MagicEyes products are not designed, intended, or authorized for use as components in systems intended for surgical implant into the body, or other applications intended to support or sustain life, or for any other application in which the failure of the MagicEyes product could create a situation where personal injury or death may occur. Should Buyer purchase or use MagicEyes products for any such unintended or unauthorized application, Buyer shall indemnify and hold MagicEyes and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that MagicEyes was negligent regarding the design or manufacture of the part.

© MagicEyes Digital

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electric or mechanical, by photocopying, recording, or otherwise, without the prior written consent of MagicEyes Digital.

HOW TO REACH US:

MagicEyes Digital co., Ltd.

463-824 #501, A-ONE Plaza, 269-6 Seohyeon-Dong, Bundang-Gu
Seongnam-City, Gyeonggi-Do, Korea
TEL: 82-31-788-0300
FAX: 82-31-707-5765

HOME PAGE:

<http://www.mesdigital.com>

DEVELOPER FORUM:

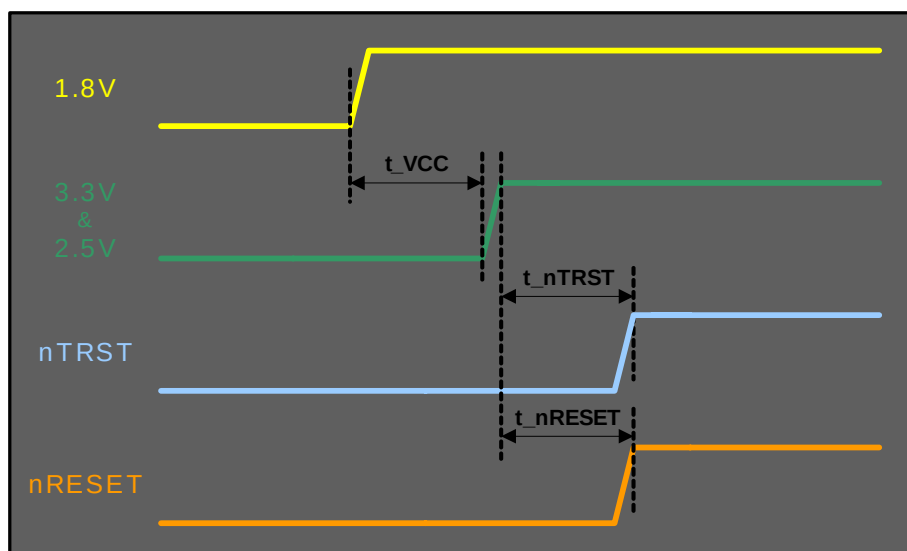
<http://forum.mesdigital.com>

Table of Contents

1. Power-ON sequence of VR3511F	4
1.1. Power circuit design guide : basic solution	4
1.1.1. Solution for PMU case	4
1.1.2. Solution for normal Power IC case	5
1.2. Generation of nTRST and nRESET	7

1. Power-ON sequence of VR3511F

<Figure 1> shows the ideal power sequence of VR3511F. Firstly, 1.8V power is supplied to VR3511F and secondly 3.3V & 2.5V power is supplied. If all powers are supplied to VR3511F simultaneously, stable boot of VR3511F can not be guaranteed.



<Figure 1. Ideal power sequence of VR3511F>

Symbol	Minimum	Typical	Maximum	Unit
t_{VCC}	10	-	-	ms
t_{nTRST}	100	-	-	ms
t_{nRESET}	100	-	-	ms

<Table 1. Power-on timing parameter specification>

1.1. Power circuit design guide : basic solution

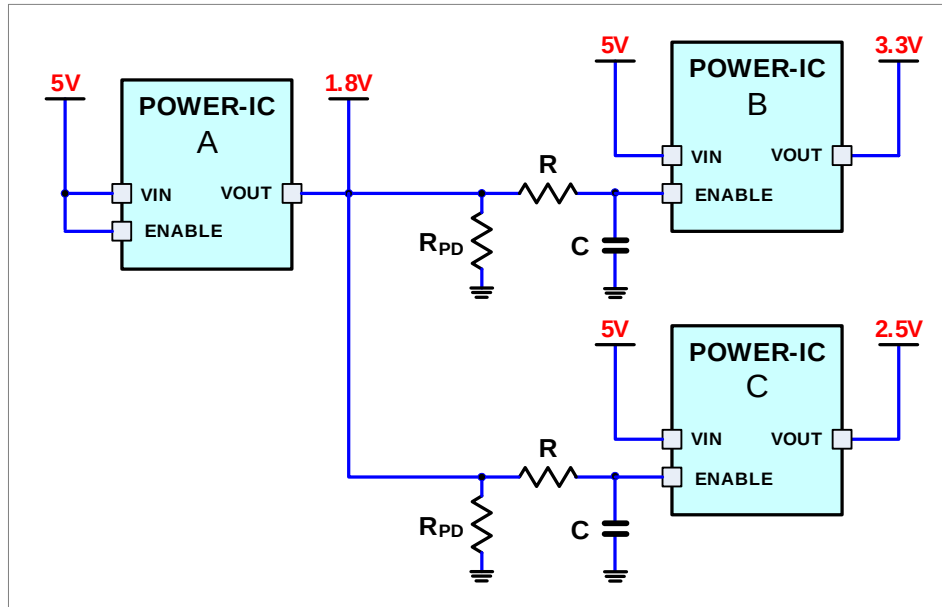
There are two solutions for stable boot. One is the use of PMU(Power Management Unit) and the other is the use of circuit which can control power sequence. **The customer of VR3511F must apply below solution to their VR3511F product.**

1.1.1. Solution for PMU case

One of solution for power circuit design is the use of PMU(but you must select the PMU which can control the power supply sequence). In this case PMU can control the power supply timing as well as the power supply sequence.

1.1.2. Solution for normal Power IC case

In case of the power circuit composed of normal power ICs, the power circuit must be designed as <Figure 2> to guarantee the stable power supply sequence.



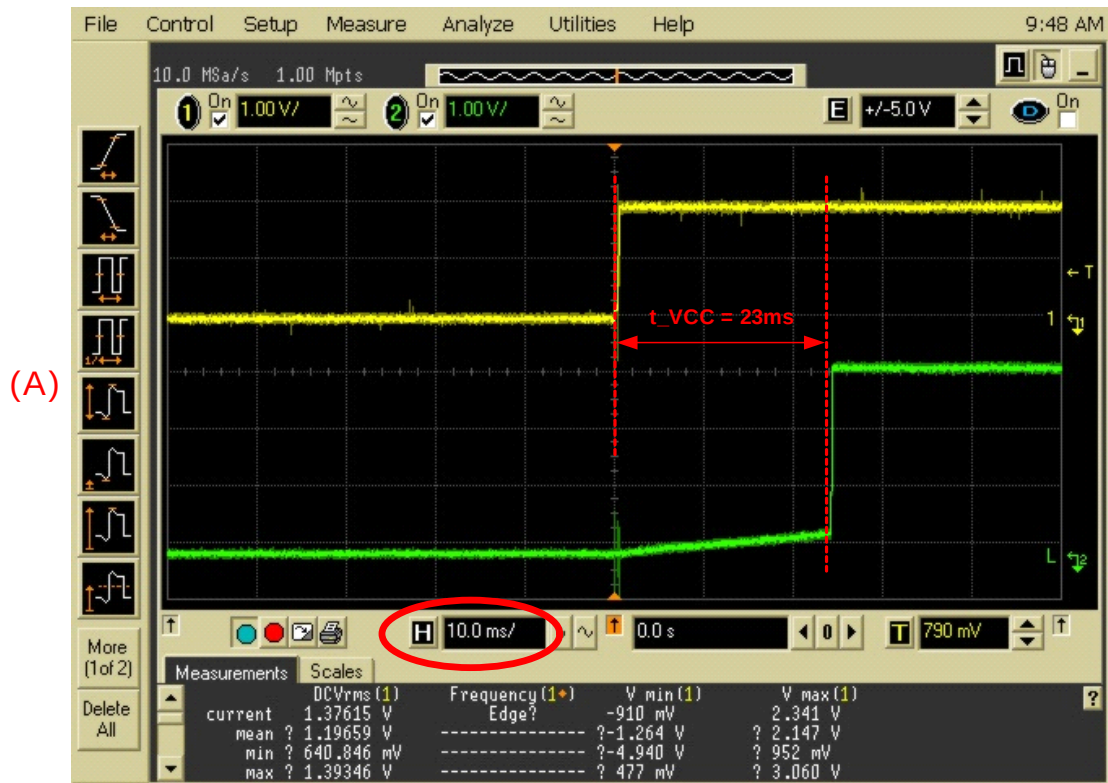
<Figure 2. Design guide for normal power circuit>

If power is supplied to board, POWER-IC A is enabled and generates 1.8V power. And then this 1.8V power enables POWER-IC B & POWER-IC C and they generate 3.3V power and 2.5V power, respectively. To use this method, POWER-IC B and POWER-IC C must have enable function pin.

<Figure 3> shows the power supply timing diagram of <Figure 2> circuit. The same as <Figure 1>, 1.8V power is supplied firstly and 3.3V power is supplied some times ago. The timing gap(t_{VCC}) between 1.8V and 3.3V can be controlled by changing the R, C value. <Table 2> shows the R, C value for <Figure 3>. R_{PD} is a Pull-Down resistor and it's value is fixed to 100Kohm.

Contents	R_{PD} (Kohm)	R (Kohm)	C (uF)
(A)	100	10	4.7
(B)	100	100	2.2

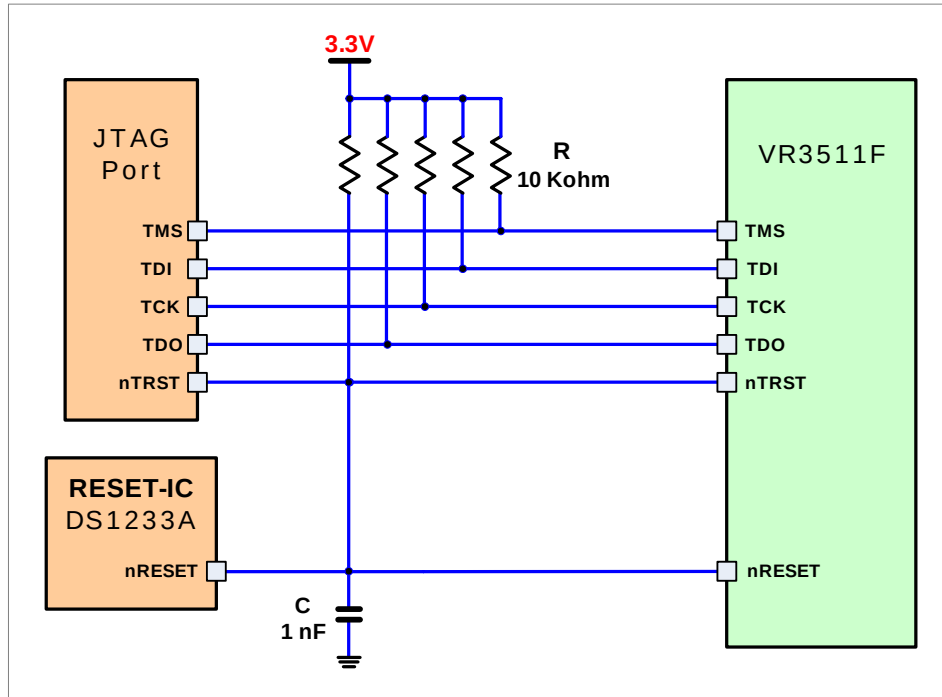
<Table 1. Power-on timing parameter specification>



<Figure 3. Timing diagram for sequential Power enable case>

1.2. Generation of nTRST and nRESET

<Figure 4> shows the circuit design for nTRST and nRESET generation. The customer of VR3511F must apply below circuit to their VR3511F product.



<Figure 4. Generation of nTRST and nRESET signal>